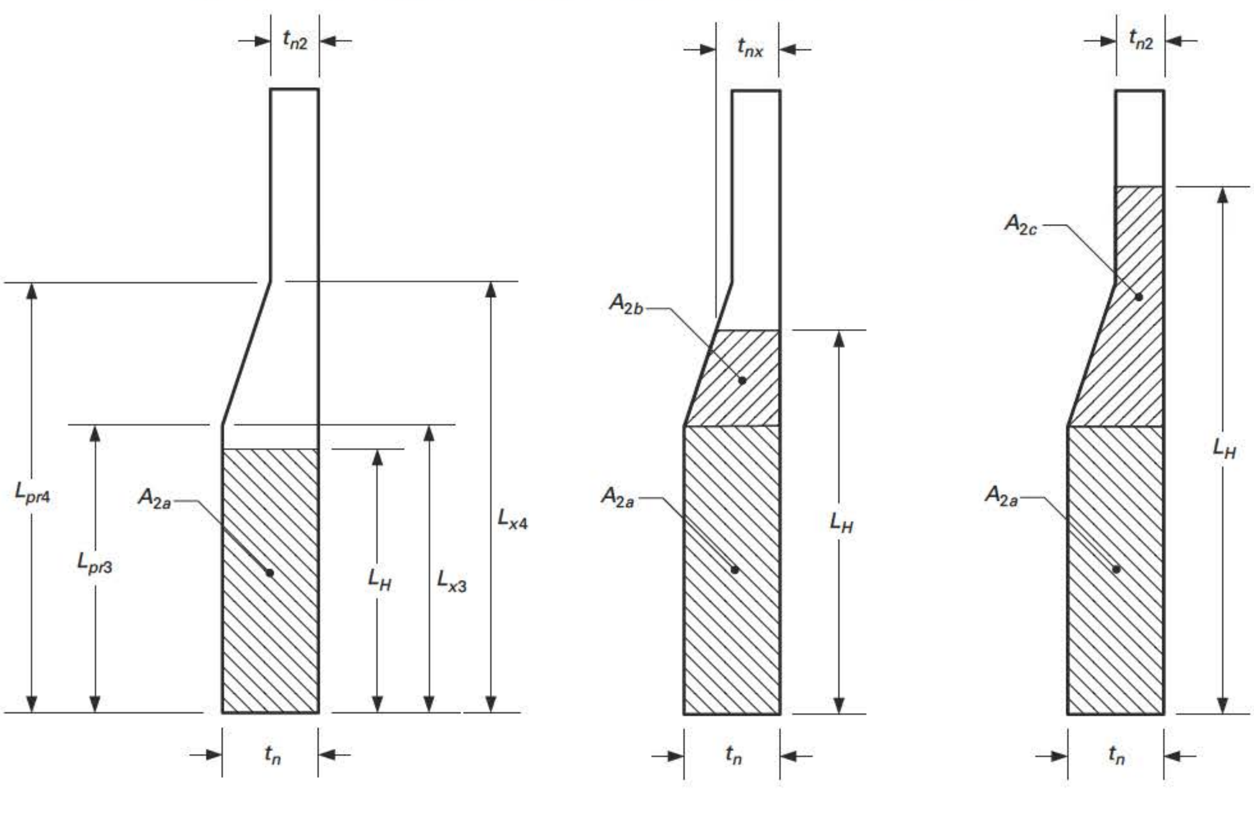
Figure 4.5.14
Metal Area Definition for A_2 With Variable Thickness of Set-on Nozzles



(a) $L_H \le L_{x3}$ (b) $L_{x3} < L_H \le L_{x4}$ (c) $L_H > L_{x4}$

4.6 DESIGN RULES FOR FLAT HEADS

4.6.1 SCOPE

4.6.1.1 The minimum thickness of unstayed flat heads, cover plates and blind flanges shall conform to the requirements given in 4.6. These requirements apply to both circular and noncircular heads and covers. Some acceptable types of flat heads and covers are shown in Table 4.6.1. In this table, the dimensions of the component parts and the dimensions of the welds are exclusive of extra metal required for corrosion allowance.

4.6.1.2 The design methods in this paragraph provide adequate strength for the design pressure. A greater thickness may be necessary if a deflection criterion is required for operation (e.g., leakage at threaded or gasketed joints).

4.6.1.3 For flat head types with a bolted flange connection where the gasket is located inside the bolt circle, calculations shall be made for two design conditions, gasket seating and operating conditions. Details regarding computation of design bolt loads for these two conditions are provided in 4.16.

4.6.2 FLAT UNSTAYED CIRCULAR HEADS

4.6.2.1 Circular blind flanges conforming to any of the flange standards listed in Part 1 and the requirements of 4.1.11 are acceptable for the diameters and pressure–temperature ratings in the respective standard when the blind flange is of the types shown in Table 4.6.1, Detail 7.

4.6.2.2 The minimum required thickness of a flat unstayed circular head or cover that is not attached with bolting that results in an edge moment shall be calculated by the following equation.

$$t = d\sqrt{\frac{CP}{S_{ho}E}} \tag{4.6.1}$$

4.6.2.3 The minimum required thickness of a flat unstayed circular head, cover, or blind flange that is attached with bolting that results in an edge moment (see Table 4.6.1, Detail 7) shall be calculated by the equations shown below. The operating and gasket seating bolt loads, W_o and W_g, and the moment arm of this load, h_G, in these equations shall be computed based on the flange geometry and gasket material as described in 4.16.

$$t = \max\left[t_o, t_g\right] \tag{4.6.2}$$

where

$$t_o = d\sqrt{\frac{CP}{S_{ho}E} + \frac{1.9W_o h_G}{S_{ho}Ed^3}} \tag{4.6.3}$$

$$t_g = d\sqrt{\frac{1.9W_g h_G}{S_{hg}Ed^3}} \tag{4.6.4}$$

4.6.3 FLAT UNSTAYED NONCIRCULAR HEADS

4.6.3.1 The minimum required thickness of a flat unstayed noncircular head or cover that is not attached with bolting that results in an edge moment shall be calculated by the following equation.

$$t = d\sqrt{\frac{ZCP}{S_{ho}E}} \tag{4.6.5}$$

where

$$Z = \min\left[2.5, \left(3.4 - \frac{2.4d}{D}\right)\right] \tag{4.6.6}$$

4.6.3.2 The minimum required thickness of a flat unstayed noncircular head, cover, or blind flange that is attached with bolting that results in an edge moment (see Table 4.6.1, Detail 7) shall be calculated by the equations shown below. The operating and gasket seating bolt loads, W_o and W_g, and the moment arm of this load, h_G, in these equations shall be computed based on the flange geometry and gasket material as described in 4.16.

$$t = \max\left[t_o, t_g\right] \tag{4.6.7}$$

where

$$t_o = d\sqrt{\frac{ZCP}{S_{ho}E} + \frac{6W_o h_G}{S_{ho}ELd^2}} \tag{4.6.8}$$

$$t_g = d\sqrt{\frac{6W_g h_G}{S_{hg}ELd^2}} \tag{4.6.9}$$

The parameter Z is given by eq. (4.6.6).

4.6.4 INTEGRAL FLAT HEAD WITH A CENTRALLY LOCATED OPENING

4.6.4.1 Flat heads which have a single, circular, centrally located opening that exceeds one-half of the head diameter shall be designed in accordance with the rules which follow. A general arrangement of an integral flat head with or without a nozzle attached at the central opening is shown in Figure 4.6.1.

(a) The shell-to-flat head juncture shall be integral, as shown in Table 4.6.1, Details 1, 2, 3, and 4. Alternatively, a butt weld, or a full penetration corner weld similar to the joints shown in Table 4.6.1 Details 5 and 6 may be used.

(b) The central opening in the flat head may have a nozzle that is integral or integrally attached by a full penetration weld, or a nozzle attached by nonintegral welds (i.e.,: a double fillet or partial penetration weld, or may have an opening without an attached nozzle or hub. In the case of a nozzle attached by nonintegral welds, the head is designed as a head without an attached nozzle or hub.

4.6.4.2 The head thickness does not have to meet the rules in 4.6.2 or 4.6.3. The flat head thickness and other geometry parameters need only satisfy the allowable stress limits in Table 4.6.3.

4.6.4.3 A procedure that can be used to design an integral flat head with a single, circular centrally located opening is shown below.

Step 1. Determine the design pressure and temperature of the flat head opening.

Step 2. Determine the geometry of the flat head opening (see Figure 4.6.1).

Step 3. Calculate the operating moment, M_o, using the following equation.

$$M_o = 0.785 B_n^2 P\left(R + \frac{g_{1n}}{2}\right) + 0.785\left(B_s^2 - B_n^2\right)P\left(\frac{R + g_{1n}}{2}\right) \tag{4.6.10}$$

where

$$R = \frac{B_s - B_n}{2} - g_{1n} \tag{4.6.11}$$

Step 4. Calculate F, V, and f based on B_n, g_{1n}, g_{0n} and h_n using the equations in Table 4.16.4 and Table 4.16.5, designate the resulting values as F_n, V_n, and f_n.

Step 5. Calculate F, V, and f based on B_s, g_{1s}, g_{0s} and h_s using the equations in Table 4.16.4 and Table 4.16.5, designate the resulting values as F_s, V_s, and f_s.

Step 6. Calculate Y, T, U, Z, L, e, and d based on $K = A/B_n$ using the equations in Table 4.16.4.

Step 7. Calculate the quantity $(E\theta)^*$ using one of the following equations:

For an opening with an integrally attached nozzle:

$$(E\theta)^* = \frac{0.91\left(\frac{g_{1n}}{g_{on}}\right)^2 \left(B_n + g_{0n}\right)V_n}{f_n\sqrt{B_n g_{0n}}} S_H \tag{4.6.12}$$

Where S_H is evaluated using the equation in Table 4.6.2.

For an opening without an attached nozzle or with a nozzle or hub attached with nonintegral welds:

$$(E\theta)^* = \frac{B_n S_T}{t} \tag{4.6.13}$$

Where S_T is evaluated using the equation in Table 4.6.2.

Step 8. Calculate the quantity M_H using the following equation:

$$M_H = \frac{(E\theta)^*}{\dfrac{1.74 V_s\sqrt{B_s g_{0s}}}{g_{0s}^3\left(B_s + g_{0s}\right)} + \dfrac{(E\theta)^*}{M_o}\left(1 + \dfrac{F_s t}{\sqrt{B_s g_{0s}}}\right)} \tag{4.6.14}$$

Step 9. Calculate the quantity X_1 using the following equation:

$$X_1 = \frac{M_o - M_H\left(1 + \dfrac{F_s t}{\sqrt{B_s g_{0s}}}\right)}{M_o} \tag{4.6.15}$$

Step 10. Calculate the stresses at the shell-to-flat head junction and opening-to-flat-head junction using Table 4.6.2.

Step 11. Check the flange stress acceptance criteria in Table 4.6.3. If the stress criteria are satisfied, then the design is complete. If the stress criteria are not satisfied, then re-proportion the flat head and/or opening dimensions and go to Step 3.

4.6.5 NOMENCLATURE

A = shell outside diameter.
B_s = inside diameter of the shell.
B_n = inside diameter of the opening.
C = factor depending upon the method of attachment of head, shell dimensions, and other items as described in Table 4.6.1.
D = is the long span of noncircular heads or covers measured perpendicular to short span.
d = diameter, or short span, measured as indicated in figure shown in Table 4.6.1.
E = joint factor.
e = flange stress factor.
f_n = hub stress correction factor for the nozzle opening-to-flat head junction.
f_s = hub stress correction factor the shell-to-flat head junction
E = weld joint factor (see 4.2).
F_n = flange stress factor for the nozzle opening-to-flat head junction.
F_s = flange stress factor for the shell-to-flat head junction
g_{1s} = hub thickness at the large end of the shell-to-flat head junction.
g_{0n} = hub thickness at the small end of the nozzle opening-to-flat head junction.
g_{0s} = hub thickness at the small end of the shell-to-flat head junction.
g_{1n} = hub thickness at the large end of the nozzle opening-to-flat head junction.
g_{1s} = hub thickness at the large end of the shell-to-flat head junction.
h_G = gasket moment arm (see Table 4.16.6).
h_n = hub length at the large end of the nozzle opening-to-flat head junction.
h_s = hub length at the large end of the shell-to-flat head junction.
L = perimeter of a noncircular bolted head measured along the centers of the bolt holes, or the flange stress factor, as applicable.
M_o = operating moment.
M_H = moment acting at the shell-to-flat head junction.
m = thickness ratio t_r/t_s.
P = internal design pressure.
r = inside corner radius on a head formed by flanging or forging.
S_{ho} = allowable stress from Annex 3-A for the head evaluated at the design temperature.
S_{hg} = allowable stress from Annex 3-A for the head evaluated at the gasket seating condition.
T = flange stress factor.
t = minimum required thickness of the flat head or cover.
t_g = required thickness of the flat head or cover for the gasket seating condition.
t_o = required thickness of the flat head or cover for the design operating condition.
t_f = nominal thickness of the flange on a forged head at the large end.
t_h = nominal thickness of the flat head or cover.
t_r = required thickness of a seamless shell.
t_s = nominal thickness of the shell.
t_1 = throat dimension of the closure weld
U = flange stress factor.
V_n = flange stress factor for the nozzle opening-to-flat head junction.
V_s = flange stress factor for the shell-to-flat head junction
W_o = operating bolt load at the design operating condition.
W_g = gasket seating bolt load at the design gasket seating condition.
Y = length of the flange of a flanged head, measured from the tangent line of knuckle, or the flange stress factor, as applicable.
Z = factor for noncircular heads and covers that depends on the ratio of short span to long span, or the flange stress factor, as applicable.
Z_1 = integral flat head stress parameter.

$(E\theta)^*$ = slope of head with central opening or nozzle times the modulus of elasticity, disregarding the interaction of the integral shell at the outside diameter of the head.

4.6.6 TABLES

(25)

Table 4.6.1
C Parameter for Flat Head Designs

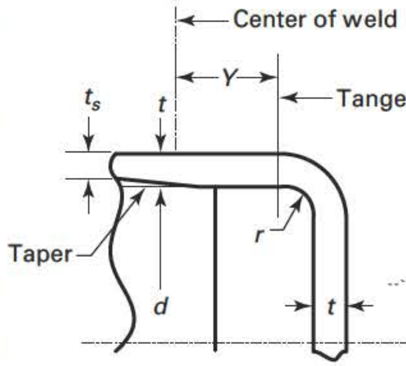
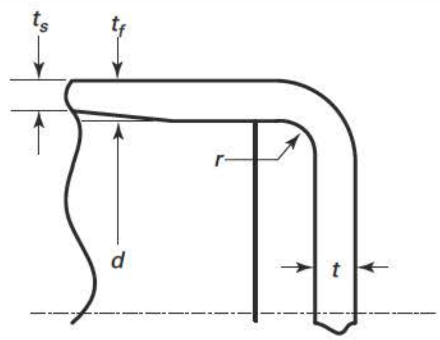
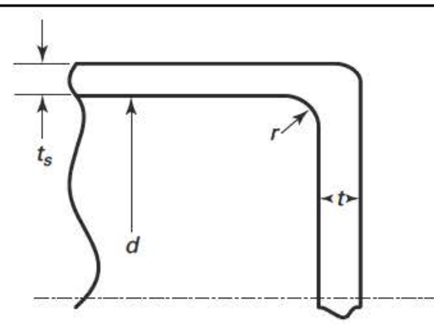
Detail	Requirements	Figure
1	• C = 0.17 for flanged circular and noncircular heads forged integral with or butt welded to the vessel with an inside corner radius not less than three times the required head thickness, with no special requirement with regard to length of flange. • C = 0.10 for circular heads, when the flange length for heads of the above design is not less than: $Y = \left[1.1 - 0.8\left(\frac{t_s}{t_h}\right)^2\right]\sqrt{dt_h}$ • C = 0.10 for circular heads, when the flange length Y less than the requirements in the above equation but the shell thickness is not less than: $t_s = 1.12t_h\sqrt{1.1 - Y/\sqrt{dt_h}}$ for a length of at least $2\sqrt{dt_s}$. When C = 0.10 is used, the taper shall be at least 1:3. • $r = 3t$ minimum shall be used	
2	• C = 0.17 for forged circular and noncircular heads integral with or butt welded to the vessel, where the flange thickness is not less than two times the shell thickness, the corner radius on the inside is not less than three times the flange thickness. • $r = 3t_f$ minimum shall be used	
3	• $C = \max[0.33m, 0.20]$ for forged circular and noncircular heads integral with or butt welded to the vessel, where the flange thickness is not less than the shell thickness, the corner radius on the inside is not less than the following: $r = 10 \text{ mm } (0.375 \text{ in.})$ for $t_s \leq 38 \text{ mm } (1.5 \text{ in.})$ $r = \min[0.25t_s, 19 \text{ mm } (0.75 \text{ in.})]$ for $t_s > 38 \text{ mm } (1.5 \text{ in.})$	

Table 4.6.1
C Parameter for Flat Head Designs (Cont'd)

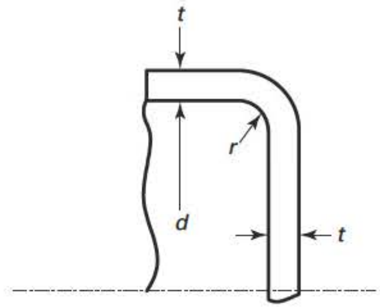
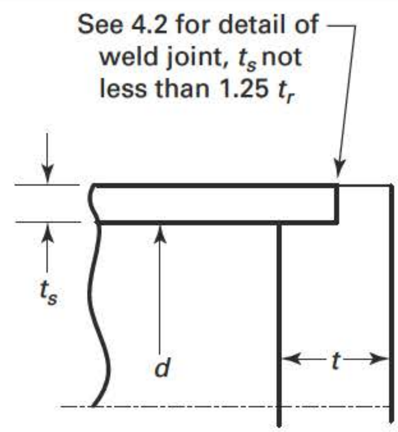
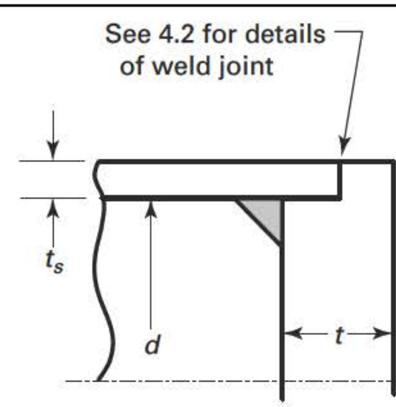
Detail	Requirements	Figure
4	• C = 0.13 for integral flat circular heads when: – the dimension d does not exceed 610 mm (24 in.) – the ratio of thickness of the head to the dimension d is not less than 0.05 or greater than 0.25 – the head thickness t_h is not less than the shell thickness t_s – the inside corner radius is not less than $0.25t$ – the construction is obtained by special techniques of upsetting and spinning the end of the shell, such as employed in closing header ends. • $r = 3t$ minimum shall be used	
5	C = 0.33 for circular plates welded to the end of the shell when t_s is at least $1.25t_r$ and the weld details conform to the requirements of 4.2.	
6	$C = \max[0.33m, 0.20]$ for circular plates if an inside fillet weld with minimum throat thickness of $0.7t_s$ is used and the details of the outside weld conform to the requirements of 4.2.	

Table 4.6.1
C Parameter for Flat Head Designs (Cont'd)

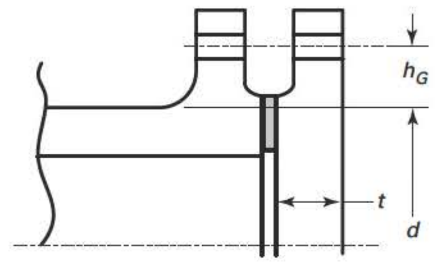
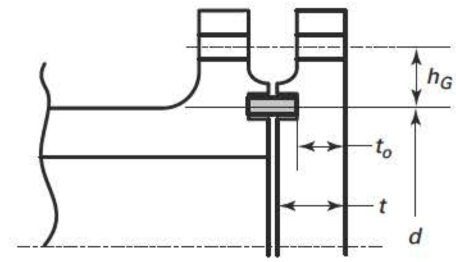
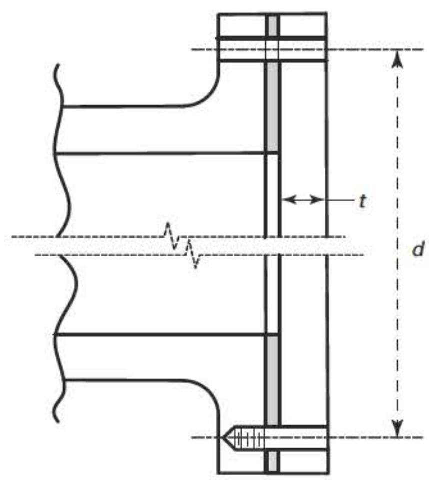
Detail	Requirements	Figure
7	• C = 0.3 for circular and noncircular heads and covers bolted to the vessel as indicated in the figures. • When the cover plate is grooved for a peripheral gasket, the net cover plate thickness under the groove or between the groove and the outer edge of the cover plate shall be not less than the following thickness. For circular heads and covers: $t_o = d\sqrt{\frac{1.9W_o h_G}{S_{ho}d^3}}$ For noncircular heads and covers: $t_o = d\sqrt{\frac{6W_o h_G}{S_{ho}Ld^2}}$	 
8	C = 0.25 for circular covers bolted with a full-face gasket to shells and flanges.	

Table 4.6.1
C Parameter for Flat Head Designs (Cont'd)

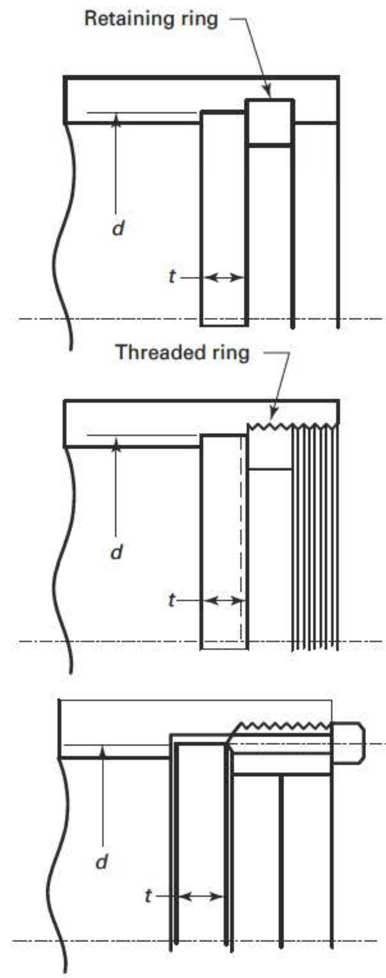
Detail	Requirements	Figure
9	C = 0.3 for a circular plate inserted into the end of a vessel and held in place by a positive mechanical locking arrangement when all possible means of failure (either by shear, tension, compression, or radial deformation, including flaring, resulting from pressure and differential thermal expansion) are designed using allowable stresses in 4.1.6. Seal welding may be used, if desired.	

Table 4.6.2
Junction Stress Equations for an Integral Flat Head With Opening

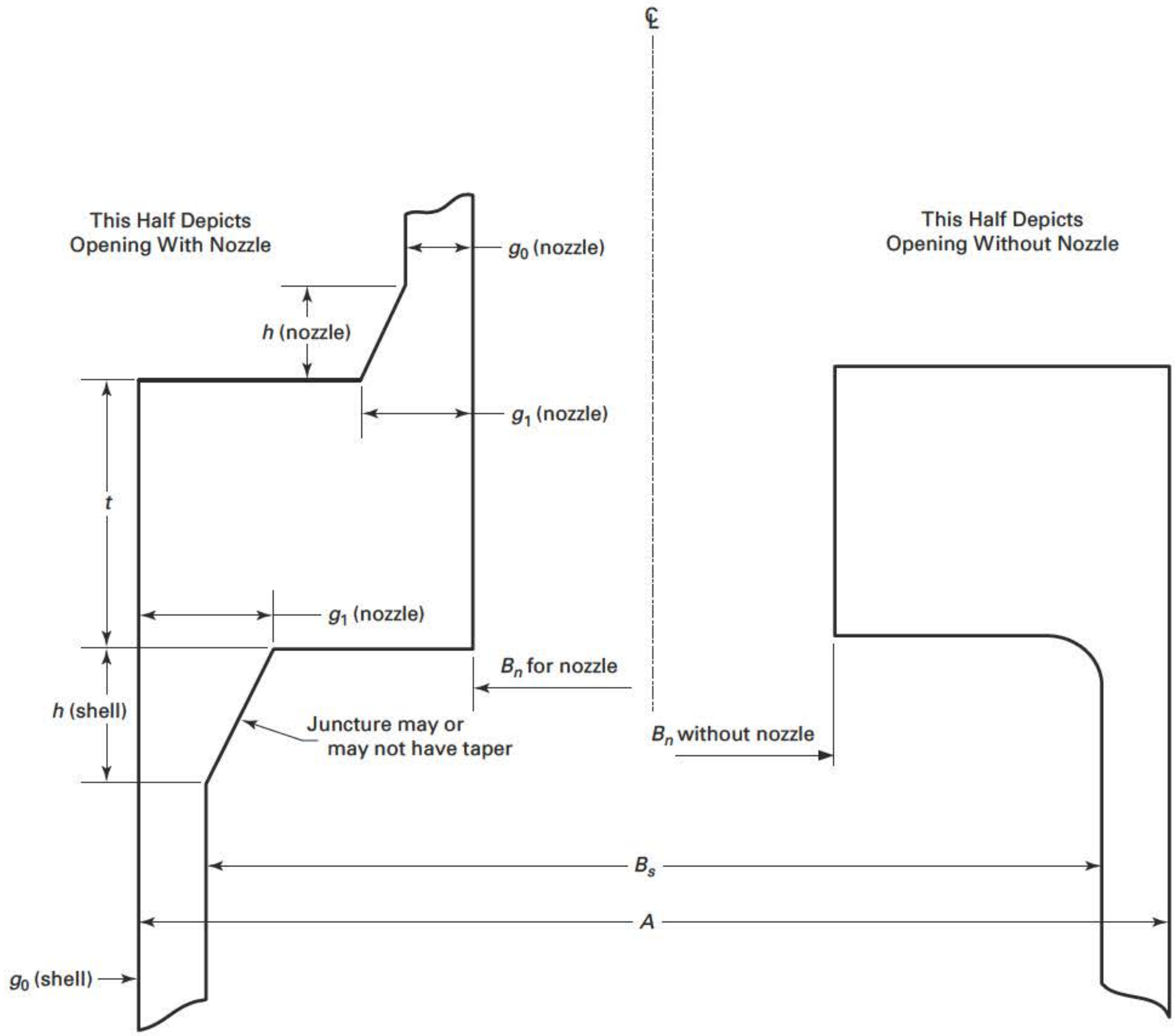
Head/Shell Junction Stresses	Opening/Head Junction Stresses
$S_{HS} = \dfrac{1.1 f_s X_1 (E\theta)^* \sqrt{B_s g_{0s}}}{\left(\dfrac{g_{1s}}{g_{0s}}\right)^2 B_s V_s}$	$S_{HO} = X_1 S_H$
$S_{RS} = \dfrac{1.91 M_H \left(1 + \dfrac{F_s t}{\sqrt{B_s g_{0s}}}\right)}{B_s t^2} + \dfrac{0.64 F_s M_H}{B_s \sqrt{B_s g_{0s}}\, t}$	$S_{RO} = X_1 S_R$
$S_{TS} = \dfrac{X_1 (E\theta)^* t}{B_s} - \dfrac{0.57 M_H \left(1 + \dfrac{F_s t}{\sqrt{B_s g_{0s}}}\right)}{B_s t^2} + \dfrac{0.64 Z F_s M_H}{B_s \sqrt{B_s g_{0s}}\, t}$	$S_{TO} = X_1 S_T + \dfrac{0.64 Z_1 F_s M_H}{B_s \sqrt{B_s g_{0s}}\, t}$
	where
	$S_H = \dfrac{f_n M_o}{L g_{1n}^2 B_n}$
	$S_R = \dfrac{(1.33te + 1) M_o}{L t^2 B_n}$
	$S_T = \dfrac{Y M_o}{t^2 B_n} - Z S_R$
	$Z_1 = \dfrac{2K^2}{K^2 - 1}$
	Note: $S_R = S_H = 0.0$ for the case of an opening without a nozzle

Table 4.6.3
Stress Acceptance Criteria for an Integral Flat Head With Opening

Head/Shell Junction Stresses	Opening/Head Junction Stresses
$S_{HS} \le 1.5 S_{ha}$	$S_{HO} \le 1.5 S_{ha}$
$S_{RS} \le S_{ha}$	$S_{RO} \le S_{ha}$
$S_{TS} \le S_{ha}$	$S_{TO} \le S_{ha}$
$\dfrac{(S_{HS} + S_{RS})}{2} \le S_{ha}$	$\dfrac{(S_{HO} + S_{RO})}{2} \le S_{ha}$
$\dfrac{(S_{HS} + S_{TS})}{2} \le S_{ha}$	$\dfrac{(S_{HO} + S_{TO})}{2} \le S_{ha}$

4.6.7 FIGURES

Figure 4.6.1
Integral Flat Head With a Large Central Opening



4.7 DESIGN RULES FOR SPHERICALLY DISHED BOLTED COVERS

4.7.1 SCOPE

4.7.1.1 Design rules for four configurations of circular spherically dished heads with bolting flanges are provided in 4.7. The four head types are shown in Figures 4.7.1, 4.7.2, 4.7.3, and 4.7.4. The design rules cover both internal and external pressure, pressure that is concave and convex to the spherical head, respectively. The maximum value of the pressure differential shall be used in all of the equations.

4.7.1.2 For head types with a bolted flange connection where the gasket is located inside the bolt circle, calculations shall be made for two design conditions, gasket seating and operating conditions. Details regarding computation of design bolt loads and flange moments for these two conditions are provided in 4.16. If a flange moment is computed as a negative number, the absolute value of this moment shall be used in all of the equations.

4.7.1.3 Calculations shall be performed using dimensions in the corroded condition and the uncorroded condition, and the more severe case shall control.

4.7.1.4 Flanges designed to Figures 4.7.2 through 4.7.5 do not need to comply with the flange rigidity criterion in Table 4.16.10.

4.7.2 TYPE A HEAD THICKNESS REQUIREMENTS

4.7.2.1 The thickness of the head and skirt for a Type A Head Configuration (see Figure 4.7.1) shall be determined in accordance with the rules in 4.3 for internal pressure (pressure on the concave side), and 4.4 for external pressure (for pressure on the convex side). The skirt thickness shall be determined using the appropriate equation for cylindrical shells. The head radius, L, and knuckle radius, r, shall comply with the limitations given in these paragraphs.

4.7.2.2 The flange thickness of the head for a Type A Head Configuration shall be determined in accordance with the rules of 4.16. When a slip-on flange conforming to the standards listed in Table 1.1 is used, design calculations per 4.16 need not be done, provided the design pressure–temperature is within the pressure–temperature rating permitted in the flange standard.

4.7.2.3 Detail (a) in Figure 4.7.1 is permitted if both of the following requirements are satisfied.

(a) The material of construction satisfies the following equation.

$$\frac{S_{yT}}{S_u} \leq 0.625 \tag{4.7.1}$$

(b) The component is not in cyclic service, i.e., a fatigue analysis is not required in accordance with 4.1.1.4.

4.7.3 TYPE B HEAD THICKNESS REQUIREMENTS

4.7.3.1 The thickness of the head for a Type B Head Configuration (see Figure 4.7.2) shall be determined by the following equations.

(a) Internal pressure (pressure on the concave side)

$$t = \frac{5PL}{6S} \tag{4.7.2}$$

(b) External pressure (pressure on the convex side) - the head thickness shall be determined in accordance with the rules in 4.4.

4.7.3.2 The flange thickness of the head for a Type B Head Configuration shall be determined by the following equations where the flange moments M_o and M_g for the operating and gasket seating conditions, respectively, are determined from 4.16.

(a) Flange thickness for a ring gasket

$$T = \max\left[T_g, T_o\right] \tag{4.7.3}$$

where

$$T_g = \sqrt{\frac{M_g}{S_{fg}B}\left(\frac{A+B}{A-B}\right)} \tag{4.7.4}$$

$$T_o = \sqrt{\frac{M_o}{S_{fo}B}\left(\frac{A+B}{A-B}\right)} \tag{4.7.5}$$

(b) Flange thickness for a full face gasket

$$T = 0.6\sqrt{\frac{|P|}{S_{fo}}\left(\frac{B(A+B)(C-B)}{A-B}\right)} \tag{4.7.6}$$

4.7.3.3 A Type B head may only be used if both of the requirements in 4.7.2.3 are satisfied.

4.7.4 TYPE C HEAD THICKNESS REQUIREMENTS

4.7.4.1 The thickness of the head for a Type C Head Configuration (see Figure 4.7.3) shall be determined by the following equations.

(a) Internal pressure (pressure on the concave side) - the head thickness shall be determined using eq. (4.7.2).

(b) External pressure (pressure on the convex side) - the head thickness shall be determined in accordance with the rules in 4.4.

4.7.4.2 The flange thickness of the head for a Type C Head Configuration shall be determined by the following equations where the flange moments M_o and M_g for the operating and gasket seating conditions, respectively, are determined from 4.16.

(a) Flange thickness for a ring gasket for heads with round bolting holes

$$T = \max\left[T_g, T_o\right] \tag{4.7.7}$$

where

$$T_g = \sqrt{\frac{1.875M_g}{S_{fg}B}\left(\frac{C+B}{7C-5B}\right)} \tag{4.7.8}$$

$$T_o = Q + \sqrt{\frac{1.875M_o}{S_{fo}B}\left(\frac{C+B}{7C-5B}\right)} \tag{4.7.9}$$

$$Q = \frac{|P|L}{4S_{fo}}\left(\frac{C+B}{7C-5B}\right) \tag{4.7.10}$$

(b) Flange thickness for ring gasket for heads with bolting holes slotted through the edge of the head

$$T = \max\left[T_g, T_o\right] \tag{4.7.11}$$

where

$$T_g = \sqrt{\frac{1.875M_g}{S_{fg}B}\left(\frac{C+B}{3C-B}\right)} \tag{4.7.12}$$

$$T_o = Q + \sqrt{\frac{1.875M_o}{S_{fo}B}\left(\frac{C+B}{3C-B}\right)} \tag{4.7.13}$$

$$Q = \frac{|P|L}{4S_{fo}}\left(\frac{C+B}{3C-B}\right) \tag{4.7.14}$$

(c) Flange thickness for full-face gasket for heads with round bolting holes

$$T = Q + \sqrt{Q^2 + \frac{3BQ(C-B)}{L}} \tag{4.7.15}$$

The parameter Q is given by eq. (4.7.10).

(d) Flange thickness for full-face gasket for heads with bolting holes slotted through the edge of the head

$$T = Q + \sqrt{Q^2 + \frac{3BQ(C-B)}{L}} \tag{4.7.16}$$

The parameter Q is given by eq. (4.7.14).

4.7.5 TYPE D HEAD THICKNESS REQUIREMENTS

4.7.5.1 The thickness of the head for a Type D Head Configuration (see Figure 4.7.4) shall be determined by the following equations.

(a) Internal pressure (pressure on the concave side) - the head thickness shall be determined using eq. (4.7.2).

(b) External pressure (pressure on the convex side) - the head thickness shall be determined in accordance with the rules in 4.4.

4.7.5.2 The flange thickness of the head for a Type D Head Configuration shall be determined by the following equations.

$$T = \max\left[T_g, T_o\right] \tag{4.7.17}$$

where

$$T_g = \sqrt{\frac{M_g}{S_{fg}B} + \left(\frac{A+B}{A-B}\right)} \tag{4.7.18}$$

$$T_o = Q + \sqrt{Q^2 + \frac{M_o}{S_{fo}B} + \left(\frac{A+B}{A-B}\right)} \tag{4.7.19}$$

$$Q = \frac{|P|\, B\sqrt{4L^2 - B^2}}{8S_{fo}(A-B)} \tag{4.7.20}$$

When determining the flange design moment for the design condition, M_o, using 4.16, the following modifications shall be made. The moment arm, h_D, shall be computed using eq. (4.7.21). An additional moment term, M_r, computed using eq. (4.7.22) shall be added to M_o as defined 4.16. The term M_{oe} in the equation for M_o as defined 4.16 shall be set to zero in this calculation. Note that this term may be positive or negative depending on the orientation of t_v, R, A_R.

$$h_D = 0.5(C - B) \tag{4.7.21}$$

$$M_r = \left(0.785B^2 P\cot[\beta]\right)h_r \tag{4.7.22}$$

where

$$\beta = \arcsin\left[\frac{B}{2L + t}\right] \tag{4.7.23}$$

4.7.5.3 As an alternative to the rules in 4.7.5.1 and 4.7.5.2, the following procedure can be used to determine the required head and flange thickness of a Type D head. This procedure accounts for the continuity between the flange ring and the head, and represents a more accurate method of analysis.

Step 1. Determine the design pressure and temperature of the flange joint. If the pressure is negative, a negative value must be used for P in all of the equations of this procedure, and

$$P_e = 0.0 \quad \text{for internal pressure} \tag{4.7.24}$$

$$P_e = P \quad \text{for external pressure} \tag{4.7.25}$$

Step 2. Determine an initial Type D head configuration geometry (see Figure 4.7.5). The following geometry parameters are required:

(a) The flange bore, B

(b) The bolt circle diameter, C

(c) The outside diameter of the flange, A

(d) Flange thickness, T

(e) Mean head radius, R

(f) Head thickness, t

(g) Inside depth of flange to the base of the head, q

Step 3. Select a gasket configuration and determine the location of the gasket reaction, G, and the design bolt loads for the gasket seating, W_g, and operating conditions, W_o, using the rules of 4.16.

Step 4. Determine the geometry parameters.

$$h_1 = \frac{(C - G)}{2} \tag{4.7.26}$$

$$h_2 = \frac{(G - B)}{2} \tag{4.7.27}$$

$$d = \frac{(A - B)}{2} \tag{4.7.28}$$

$$n = \frac{T}{t} \tag{4.7.29}$$

$$K = \frac{A}{B} \tag{4.7.30}$$

$$\phi = \arcsin\left[\frac{B}{2R}\right] \tag{4.7.31}$$

$$e = q - \frac{1}{2}\left(T - \frac{t}{\cos[\phi]}\right) \tag{4.7.32}$$

$$k_1 = 1 - \left(\frac{1 - 2v}{2\lambda}\right)\cot[\phi] \tag{4.7.33}$$

$$k_2 = 1 - \left(\frac{1 + 2v}{2\lambda}\right)\cot[\phi] \tag{4.7.34}$$

$$\lambda = \left[3\left(1 - v^2\right)\left(\frac{R}{t}\right)^2\right]^{0.25} \tag{4.7.35}$$

Step 5. Determine the shell discontinuity geometry factors.

$$C_1 = \frac{0.275n^3 t \cdot \ln[K]}{k_1} - e \tag{4.7.36}$$

$$C_2 = \frac{1.1\lambda n^3 t \cdot \ln[K]}{Bk_1} + 1 \tag{4.7.37}$$

$$C_4 = \frac{\lambda \sin[\phi]}{2}\left(k_2 + \frac{1}{k_1}\right) + \frac{B}{4nd} + \frac{1.65e}{tk_1} \tag{4.7.38}$$

$$C_5 = \frac{1.65}{tk_1}\left(1 + \frac{4\lambda e}{B}\right) \tag{4.7.39}$$

Step 6. Determine the shell discontinuity load factors for the operating and gasket conditions.

$$C_{3o} = \frac{\pi B^2 P}{4}\left(e \cdot \cot\left[\phi\right] + \frac{2q(T-q)}{B} - h_2\right) - W_o\, h_1 \tag{4.7.40}$$

$$C_{6o} = \frac{\pi B^2 P}{4}\left(\frac{4q - B \cdot \cot\left[\phi\right]}{4nd} - \frac{0.35}{\sin\left[\phi\right]}\right) \tag{4.7.41}$$

$$C_{3g} = -W_g h_1 \tag{4.7.42}$$

$$C_{6g} = 0.0 \tag{4.7.43}$$

Step 7. Determine the shell discontinuity force and moment for the operating and gasket conditions.

$$V_{do} = \frac{C_2 C_{6o} - C_{3o} C_5}{C_2 C_4 - C_1 C_5} \tag{4.7.44}$$

$$M_{do} = \frac{C_1 C_{6o} - C_{3o} C_4}{C_2 C_4 - C_1 C_5} \tag{4.7.45}$$

$$V_{dg} = \frac{C_2 C_{6g} - C_{3g} C_5}{C_2 C_4 - C_1 C_5} \tag{4.7.46}$$

$$M_{dg} = \frac{C_1 C_{6g} - C_{3g} C_4}{C_2 C_4 - C_1 C_5} \tag{4.7.47}$$

Step 8. Calculate the stresses in the head and at the head-to-flange junction using Table 4.7.1 and check the stress acceptance criteria. If the stress criteria are satisfied, then the design is complete. If the stress criteria are not satisfied, then re-proportion the bolted head dimensions and go to Step 3.

4.7.6 NOMENCLATURE

A = flange outside diameter
B = flange inside diameter
C = bolt circle diameter
C_1 = shell discontinuity geometry parameter for the Type D head alternative design procedure
C_2 = shell discontinuity geometry parameter for the Type D head alternative design procedure
C_{3g} = shell discontinuity load factor for the gasket seating condition for the Type D head alternative design procedure
C_{3o} = shell discontinuity load factor for the design operating condition for the Type D head alternative design procedure
C_4 = shell discontinuity geometry parameter for the Type D head alternative design procedure
C_5 = shell discontinuity geometry parameter for the Type D head alternative design procedure
C_{6g} = shell discontinuity load factor for the gasket seating condition for the Type D head alternative design procedure
C_{6o} = shell discontinuity load factor for the design operating condition for the Type D head alternative design procedure
e = geometry parameter for the Type D head alternative design procedure
h_1 = geometry parameter for the Type D head alternative design procedure
h_2 = geometry parameter for the Type D head alternative design procedure
h_r = moment arm of the head reaction force
K = geometry parameter for the Type D head alternative design procedure
k_1 = geometry parameter for the Type D head alternative design procedure

k_2 = geometry parameter for the Type D head alternative design procedure
L = inside crown radius
M_{dg} = shell discontinuity moment for the gasket seating condition
M_{do} = shell discontinuity moment for design operating condition
M_g = flange design moment for the gasket seating condition determined using 4.16
M_o = flange design moment for the design condition determined using 4.16 (see 4.7.5.2 for exception)
M_r = moment from the head reaction force
n = geometry parameter for the Type D head alternative design procedure
P = design pressure
P_e = pressure factor to adjust the design rules for external pressure
q = inside depth of the flange to the base of the head
R = mean radius of a Type D head
r = inside knuckle radius. S_{fg} allowable stress from Annex 3-A for the flange evaluated at the gasket seating condition
S_{fm} = membrane stress in the flange
S_{fmbi} = membrane plus bending stress on the inside surface of the flange
S_{fmbo} = membrane plus bending stress on the outside surface of the flange
S_{fo} = allowable stress from Annex 3-A for the flange evaluated at the design temperature
S_{hb} = bending stress at the head-to-flange junction
S_{hg} = allowable stress from Annex 3-A for the head evaluated at the gasket seating condition
S_{hl} = local membrane stress at the head-to-flange junction
S_{hlbi} = local membrane plus bending stress at the head-to-flange junction on the inside surface of the head
S_{hlbo} = local membrane plus bending stress at the head-to-flange junction on the outside surface of the head
S_{hm} = head membrane stress
S_{ho} = allowable stress from Annex 3-A for the head evaluated at the design temperature
S_u = minimum specified ultimate tensile strength from Annex 3-D
S_{yT} = yield strength from Annex 3-D evaluated at the design temperature
T = flange thickness
t = required head thickness
T_g = required flange thickness for the gasket seating condition
T_o = required flange thickness for design operating condition
T^* = flange thickness for a Type C head
V_{dg} = shell discontinuity shear force for the gasket seating condition
V_{do} = shell discontinuity shear force for design operating condition
W_g = bolt load for the gasket seating condition
W_o = bolt load for design operating condition
β = angle formed by the tangent to the center line of the dished cover thickness at its point of intersection with the flange ring, and a line perpendicular to the axis of the dished cover
λ = geometry parameter for the Type D head alternative design procedure
ν = Poisson's ratio
ϕ = one-half central angle of the head for the Type D head alternative design procedure

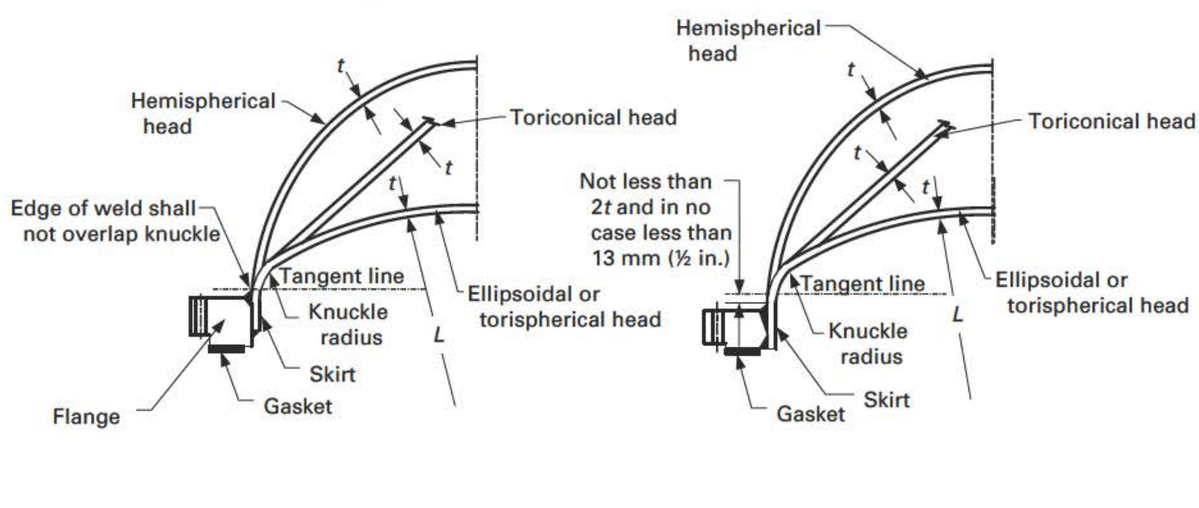
4.7.7 TABLES

Table 4.7.1
Junction Stress Equations and Acceptance Criteria for a Type D Head

Operating Conditions	Gasket Seating Conditions
$S_{hm} = \frac{PR}{2t} + P_e$	$S_{hm} = 0.0$
$S_{hl} = \frac{PR}{2t} + \frac{V_{do}\cos[\phi]}{\pi Bt} + P_e$	$S_{hl} = \frac{V_{dg}\cos[\phi]}{\pi Bt}$
$S_{hb} = \frac{6M_{do}}{\pi Bt^2}$	$S_{hb} = \frac{6M_{dg}}{\pi Bt^2}$
$S_{hlbi} = S_{hl} - S_{hb}$	$S_{hlbi} = S_{hl} - S_{hb}$
$S_{hlbo} = S_{hl} + S_{hb}$	$S_{hlbo} = S_{hl} + S_{hb}$
$S_{fm} = \frac{1}{\pi BT}\left(\frac{\pi B^2 P}{4}\left(\frac{4q}{B} - \cot[\phi]\right) - V_{do}\left(\frac{K^2+1}{K^2-1}\right)\right) + P_e$	$S_{fm} = \frac{1}{\pi BT}\left(-V_{dg}\left(\frac{K^2+1}{K^2-1}\right)\right)$
$S_{fb} = \frac{0.525n}{Btk_1}\left(V_{do} - \frac{4M_{do}\lambda}{B}\right)$	$S_{fb} = \frac{0.525n}{Btk_1}\left(V_{dg} - \frac{4M_{dg}\lambda}{B}\right)$
$S_{fmbo} = S_{fm} - S_{fb}$	$S_{fmbo} = S_{fm} - S_{fb}$
$S_{fmbi} = S_{fm} + S_{fb}$	$S_{fmbi} = S_{fm} + S_{fb}$
Acceptance Criteria	
$S_{hm} \le S_{ho}$	$S_{hm} \le S_{hg}$
$S_{hl} \le 1.5S_{ho}$	$S_{hl} \le 1.5S_{hg}$
$S_{hlbi} \le 1.5S_{ho}$	$S_{hlbi} \le 1.5S_{hg}$
$S_{hlbo} \le 1.5S_{ho}$	$S_{hlbo} \le 1.5S_{hg}$
$S_{fm} \le S_{fo}$	$S_{fm} \le S_{fg}$
$S_{fmbo} \le 1.5S_{fo}$	$S_{fmbo} \le 1.5S_{fg}$
$S_{fmbi} \le 1.5S_{fo}$	$S_{fmbi} \le 1.5S_{fg}$

4.7.8 FIGURES

Figure 4.7.1
Type A Dished Cover With a Bolting Flange



(a) Loose Flange Type

(b) Integral Flange Type

GENERAL NOTE: See Table 4.2.5, Details 2 and 3 for transition requirements for a head and skirt with different thicknesses.

Figure 4.7.2
Type B Spherically Dished Cover With a Bolting Flange

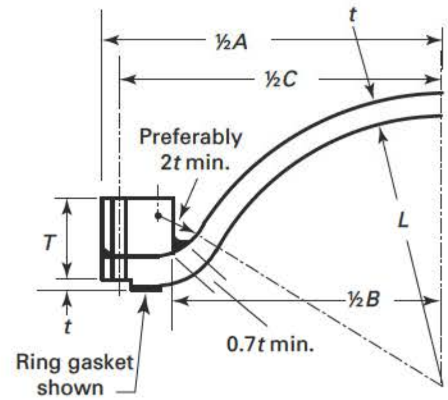
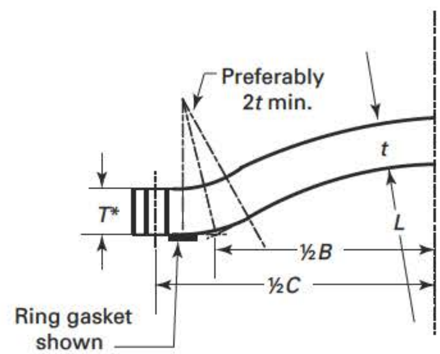


Figure 4.7.3
Type C Spherically Dished Cover With a Bolting Flange



$T^* = T \geq t$

Figure 4.7.4
Type D Spherically Dished Cover With a Bolting Flange

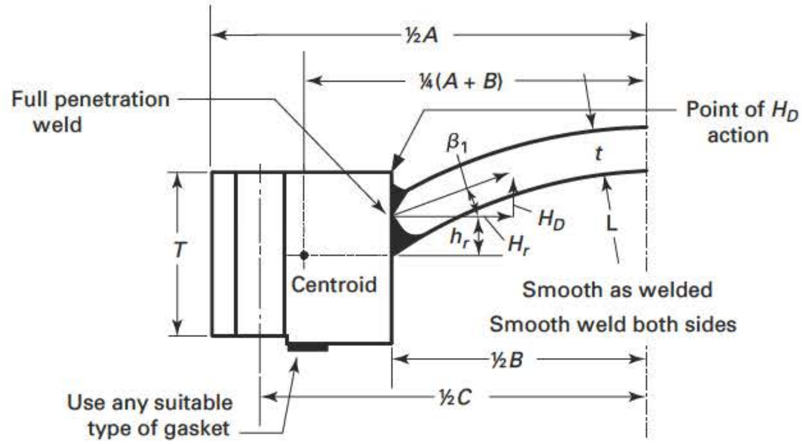
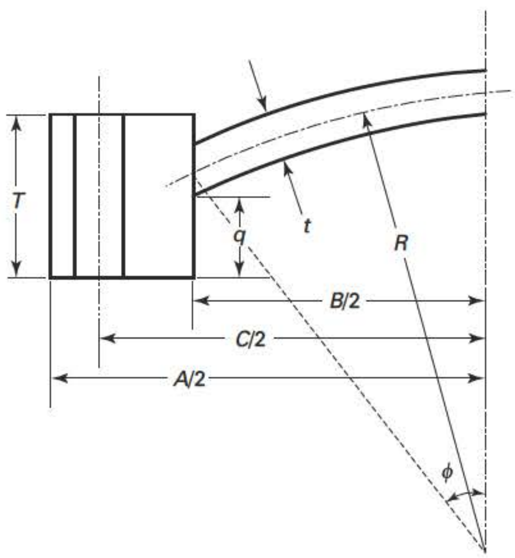


Figure 4.7.5
Type D Head Geometry for Alternative Design Procedure



(25) 4.8 DESIGN RULES FOR QUICK-OPENING AND QUICK-ACTUATING CLOSURES

4.8.1 SCOPE

4.8.1.1 Design requirements for quick-opening and quick-actuating closures are provided in 4.8. Specific calculation methods are not provided in Part 4. The rules of Part 5 shall be used to qualify the design requirements specific to quick-opening and quick-actuating closures that are not covered in Part 4.

4.8.2 DEFINITIONS

4.8.2.1 Quick-opening closures are closures other than

(a) bolted flange joints as described in 4.1.11
(b) bolted flange joints meeting the requirements of 4.16
(c) dished covers with a bolting flange meeting the requirements of 4.7
(d) clamped connections meeting the requirements of 4.17

(e) closures with multiple, manually operated swing bolts
(f) quick-actuating closures as described in 4.8.2.2

4.8.2.2 Quick-actuating closures are closures that are operated by an action that releases all holding elements.

4.8.2.3 Holding elements are parts of the closure used to hold the cover to the vessel or to provide the load required to seal the closure.

4.8.2.4 Locking elements are parts of the closure that prevent a reduction in the load on a holding element or prevent the release of a holding element. Locking elements may also be used as holding elements.

4.8.2.5 The locking mechanism or locking device may consist of a combination of locking elements.

4.8.3 GENERAL REQUIREMENTS

(a) Closures shall be designed and installed such that it may be determined by visual external observation that the holding elements are in satisfactory condition.
(b) If clamps used in the design of quick-actuating closures meet the scope of 4.17, then the requirements of 4.17 shall also be met.
(c) When a closure is provided as a part, it shall be provided with a Partial Data Report as meeting the applicable requirements.
(d) Annex 4-B provides supplementary information for the Manufacturer of the closure and provides guidance on installation, operation, and maintenance for the owner and user.
(e) The design rules of 4.16 of this Division may not be applicable to the design of closures; see 4.16.1.4.
(f) The design shall consider the effects of cyclic and other loadings [see 2.2.3.1(f)] and mechanical wear on the holding and locking elements and sealing surfaces.
(g) The Manufacturer of a pressure vessel with a quick-actuating or quick-opening closure shall supply the user with an installation, operation, and maintenance manual that shall address the maintenance and operation of the closure. The manual should address the topics discussed in Annex 4-B. The intent is for this manual to stay with the owner or operator of the pressure vessel.
(h) The use of a multilink component, such as a chain, as a holding or locking element is not permitted.
(i) Hinge pins or bolts may be used as holding elements.

4.8.4 QUICK-OPENING CLOSURES

(a) Quick-opening closures shall be designed such that the failure of a single holding element while the vessel is pressurized (or contains a static head of liquid acting at the closure) will not
(1) cause or allow the closure to be opened or leak
(2) increase the stress in any other holding element by more than 50% above the allowable stress of the element
(b) All vessels having quick-opening closures shall have a pressure release device (e.g., vent valve, threaded plug) installed on the vessel that will relieve the pressure inside the vessel prior to opening the closure. Alternatively, if release of the product in the vessel could be dangerous to personnel or the environment or could cause other safety issues, the provisions for pressure release need not be furnished when operating procedures are such that they can ensure there is no pressure in the vessel prior to opening the closure.

4.8.5 QUICK-ACTUATING CLOSURES

(a) Quick-actuating closures shall be designed such that the failure of a single locking element while the vessel is pressurized (or contains a static head of liquid acting at the closure) will not
(1) cause or allow the closure to be opened or leak
(2) result in the failure of any other locking element or holding element
(3) increase the stress in any other locking element or holding element by more than 50% above the allowable stress of the element
(b) Quick-actuating closures shall be designed such that the locking elements are engaged prior to or upon application of pressure and remain engaged until the pressure is released.
(c) Quick-actuating closures shall be designed so that all locking elements can be verified to be fully engaged by visual observation or other means prior to the application of pressure to the vessel.
(d) When installed, all vessels having quick-actuating closures shall be provided with a pressure-indicating device visible from the operating area and suitable for detecting pressure at the closure.
(e) Quick-actuating closures that are held in position by positive locking devices and that are fully released by partial rotation or limited movement of the closure itself or the locking mechanism, and any closure that is other than manually operated, shall be so designed that when the vessel is installed the following conditions are met (see also Annex 4-B):

(1) The closure and its holding elements are fully engaged in their intended operating position before pressure can be applied in the vessel.

(2) Pressure tending to force the closure open or discharge the vessel contents clear of the vessel shall be released before the closure can be fully opened for access.

(3) In the event that compliance with (1) and (2) is not inherent in the design of the closure and its holding elements, provisions shall be made so that devices to accomplish this can be added when the vessel is installed.

(f) It is recognized that it is impractical to write requirements to cover the multiplicity of quick-actuating closures or to prevent negligent operation or the circumventing of safety devices. Any device or devices that will provide the safeguards described in (e) will meet the intent of these rules.

(g) Quick-actuating closures that are held in position by a locking mechanism designed for manual operation shall be designed such that if an attempt is made to open the closure when the vessel is under pressure, the closure will leak prior to full disengagement of the locking elements and release of the closure. The design of the closure and vessel shall be such that any leakage shall be directed away from the normal position of the operator.

(h) Manually operated closures need not satisfy (e), but pressure vessels equipped with such closures shall be equipped with an audible or visible warning device that will warn the operator if pressure is applied to the vessel before the holding elements and locking elements are fully engaged in their intended position or if an attempt is made to disengage the locking mechanism before the pressure within the vessel is released.

4.9 DESIGN RULES FOR BRACED AND STAYED SURFACES

4.9.1 SCOPE

4.9.1.1 Design requirements for braced and stayed surfaces are provided in this paragraph. Requirements for the plate thickness and requirements for the staybolt or stay geometry including size, pitch, and attachment details are provided.

4.9.2 REQUIRED THICKNESS OF BRACED AND STAYED SURFACES

4.9.2.1 The minimum thickness for braced and stayed flat plates and those parts that, by these rules, require staying as flat plates with braces or staybolts of uniform diameter symmetrically spaced, shall be calculated by the following equation.

$$t = p_s\sqrt{\frac{P}{SC}} \tag{4.9.1}$$

4.9.2.2 When stays are used to connect two plates, and only one of these plates requires staying, the value of C shall be governed by the thickness of the plate requiring staying.

4.9.3 REQUIRED DIMENSIONS AND LAYOUT OF STAYBOLTS AND STAYS

4.9.3.1 The required area of a staybolt or stay at its minimum cross section, usually located at the root of the thread, exclusive of any corrosion allowance, shall be obtained by dividing the load on the staybolt computed in accordance with 4.9.3.2 by the allowable tensile stress value for the staybolt material, multiplying the result by 1.10.

4.9.3.2 The area supported by a staybolt or stay shall be computed on the basis of the full pitch dimensions, with a deduction for the area occupied by the stay. The load carried by a stay is the product of the area supported by the stay and the maximum allowable working pressure. When a staybolt or stay for a shell is unsymmetrical because of interference with other construction details, the area supported by the staybolt or stay shall be computed by taking the distance from the center of the spacing on one side of the staybolt or stay to the center of the spacing on the other side.

4.9.3.3 When the edge of a flat stayed plate is flanged, the distance from the center of the outermost stays to the inside of the supporting flange shall not be greater than the pitch of the stays plus the inside radius of the flange.

4.9.4 REQUIREMENTS FOR WELDED-IN STAYBOLTS AND WELDED STAYS

4.9.4.1 Welded-in staybolts may be used, provided the following requirements are satisfied.

(a) The configuration is in accordance with the typical arrangements shown in Figure 4.9.1.

(b) The required thickness of the plate shall not exceed 38 mm (1.5 in.).

(c) The maximum pitch shall not exceed 15 times the diameter of the staybolt; however, if the required plate thickness is greater than 19 mm (0.75 in.), the staybolt pitch shall not exceed 508 mm (20 in.).

(d) The size of the attachment welds is not less than that shown in Figure 4.9.1.

(e) The allowable load on the welds shall not exceed the product of the weld area (based on the weld dimension parallel to the staybolt), the allowable tensile stress of the material being welded, and a weld joint factor of 60%.

4.9.4.2 Welded stays may be used, provided the following requirements are satisfied.

(a) The configuration is in accordance with the typical arrangements shown in Figure 4.9.1.

(b) The pressure does not exceed 2 MPa (300 psi).

(c) The required thickness of the plate does not exceed 13 mm (0.5 in.).

(d) The size of the fillet welds is not less than the plate thickness requiring stay.

(e) The inside welds are visually examined before the closing plates are attached.

(f) The allowable load on the fillet welds shall not exceed the product of the weld area (based on the minimum leg dimension), the allowable tensile stress of the material being welded, and a weld joint factor of 55%.

(g) The maximum diameter or width of the hole in the plate shall not exceed 32 mm (1.25 in.).

(h) The maximum pitch, p_s, is determined by eq. (4.9.1) with C= 2.1 if either plate thickness is less than or equal to 11 mm (0.4375 in.) thick, and C= 2.2 for all other plate thicknesses.

4.9.5 NOMENCLATURE

C = stress factor for braced and stayed surfaces (see Table 4.9.1)
P = design pressure
P_s = maximum pitch. The maximum pitch is the greatest distance between any set of parallel straight lines passing through the centers of staybolts in adjacent rows. Each of the three parallel sets running in the horizontal, the vertical, and the inclined planes shall be considered.
S = allowable stress from Annex 3-A evaluated at the design temperature
t = minimum required plate thickness
t_s = nominal thickness of the thinner stayed plate (see Figure 4.9.1)

4.9.6 TABLES

Table 4.9.1
Stress Factor for Braced and Stayed Surfaces

Braced and Stayed Surface Construction	C
Welded stays through plates not over 11 mm (0.4375 in.) in thickness	2.1
Welded stays through plates over 11 mm (0.4375 in.) in thickness	2.2